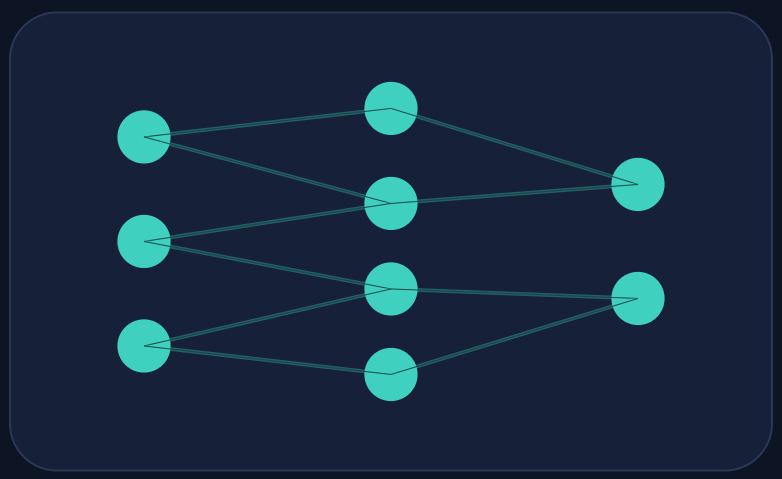
EVOLUTIONARY AI × AGENTIC AI

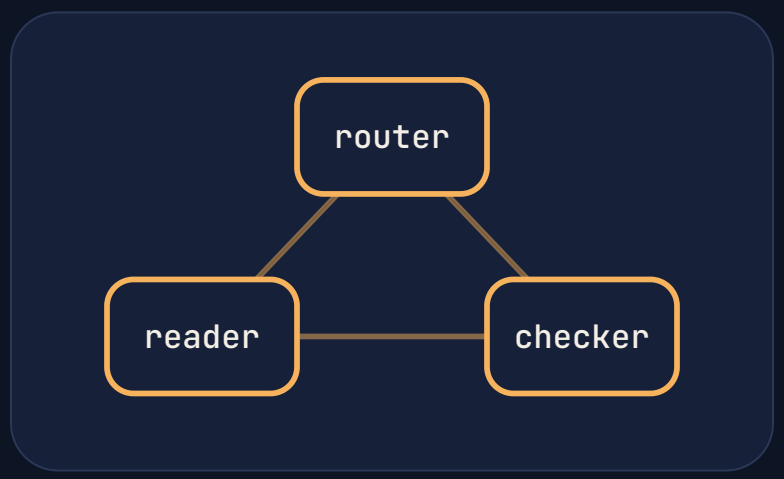
Designing agent networks was solved before agents were popular.

What two Cognizant AI Lab papers from 2019 and 2020 taught me about building multi-agent systems.

2019 · evolve layers



today · evolve agents



Siva Kumar Raju · neuro-san-esp

Swipe →

THE QUESTION CHANGED

The hard design question in AI has moved up a level.

Then

How many **layers**? How do they connect? How big is each one?

Now

How many **agents**? Who routes to whom? Which model does each one run?

Most teams answer the new question by intuition: add another agent, try a bigger model, ship it.

But the old question already has a well-tested answer.

IDEA 1 · 2019

LEAF · Evolutionary Neural AutoML · GECCO 2019

Evolve the architecture, not just the settings.

Cognizant AI Lab's LEAF evolves a neural network's structure: how many layers, how they connect, how big each one is.

The finding that stayed with me

Changing the architecture beat tuning hyperparameters, and the evolved networks were often smaller.

IDEA 2 · 2020

ESP · Surrogate-assisted Prescription · GECCO 2020

When every real test is expensive, predict first.

A

Measure a few options for real

real cost

B

Train a cheap predictor on them

seconds, no model calls

C

Screen thousands of new options against it

free

D

Pay to measure only the best few, then retrain

real cost, small

The expensive step stays small. The search can still be wide.

PUT THEM TOGETHER

LEAF
evolve the structure

+

ESP
predict before paying

=

agents
instead of layers

neuro-san-esp, built on Neuro SAN, Cognizant AI Lab's open-source multi-agent framework

→ Each candidate is a real Neuro SAN agent network, plus the model each agent runs

→ Seven mutations change it:

`add` `remove` `split` `merge` `rewire` `toggle search` `change model`

→ A predictor screens candidates for free. Only the best are measured on real model calls.

SAME IDEAS, OPPOSITE DIRECTION

ESP was already inside Neuro SAN. I pointed it the other way.

Inside Neuro SAN today

An agent network uses ESP to help a person make a decision.

Agent network uses ESP → **a person's decision**

In neuro-san-esp

ESP is used to design the agent network itself.

ESP designs → **the agent network**

LEAF'S FINDING, ONE LEVEL UP

The smaller network won.

Two-agent network — **94.5%**

stronger model on the agent that reads documents

Four-agent design — **84.0%**

every agent on the cheaper model

−37%
tokens per question

2 vs 4
agents

p=0.0007
sign test
17 wins vs 2

100 questions the networks were never tuned on, each asked twice (200 answers per network). Every run is committed in the repo and recomputed by a test.

WHAT THIS DOES NOT PROVE

Three honest limits.

It costs more per answer

The stronger model on the reader made each question cost 71% more, even with fewer tokens.

Structure vs model is not separated yet

The four-agent design was not tested with its own reader upgraded. That is the next experiment.

I’m not the first to search agent designs

My contribution is narrower: a working way to measure and compare Neuro SAN networks, which didn't exist before.

THE BIGGER POINT

The research came first. The use case has finally caught up.

Evolutionary AI built to design neural networks is useful again, now that we build systems out of agents instead of layers.

How is your team deciding how many agents to use, and which model each one gets?

Tell me in the comments.

ESP · Francon, Gonzalez, Hodjat, Meyerson, Miikkulainen, Qiu, Shahrzad · GECCO 2020
LEAF · Liang, Meyerson, Hodjat, Fink, Mutch, Miikkulainen · GECCO 2019
Code · github.com/Sivakumarraj/neuro-san-esp